CORTEX — Post-Market Surveillance (PMS) Module

Module Specification v1.2 — February 2026

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1. Overview

The PMS module manages the post-market lifecycle of clinical evidence for a device. It coordinates three complementary activities:

1. **PMS Plan** — the surveillance strategy document defining what data to collect, how, and when
2. **PMCF (Post-Market Clinical Follow-up)** — the clinical component: surveys, literature updates, gap tracking
3. **PSUR (Periodic Safety Update Report)** — the annual report aggregating all PMS data and concluding on benefit-risk acceptability

The PMS module closes the regulatory loop: gaps identified during SOA and Validation feed into PMCF actions, which generate new data, which feeds into the PSUR, which may trigger a CER update.

Under MDR, the **PSUR is the sole comprehensive post-market report**. There is no separate "PMS Report" — the PSUR aggregates both clinical (PMCF) and non-clinical (complaints, vigilance, maintenance) post-market data. The **PMCF Report** is an independent clinical follow-up document that feeds into the PSUR as input.

Real-World Evidence (RWE) — including field performance data from surveys, complaint analysis, and post-market literature — is managed exclusively within the PMS module. Pre-market clinical evaluation (SOA, Validation, CER) relies on published literature and controlled validation studies.

Vigilance reporting timelines, PMS activity periodicity, and escalation procedures are defined in Avicenna.AI's QMS procedures (AV-P17 Vigilance Management, AV-P19 NC & Complaints Management). CORTEX provides the data collection and reporting infrastructure but does not enforce or calculate regulatory deadlines. Timeline compliance is managed through the QMS.

Core principles:

- PMS is a continuous cycle, not a one-time activity
- Every PMS cycle is traceable and feeds the regulatory file
- PMCF activities are planned and tracked; results are structured for PSUR consumption
- The PSUR concludes with a benefit-risk re-assessment that determines whether a CER update is needed

```
SOA (gaps) ──┐
Validation   │
(fairness    ├──▶│ PMS Plan  │──▶│ PMCF + PMS  │──▶│ PSUR  │
 gaps)       │  │ (strategy) │  │ Activities  │  │ (annual  │
CER (open    │  └──────────┘  └──────────┘  │ report) │
questions) ──┘                 │      └──────┬──────┘
                               │
                               ▼          ▼
                        ┌──────────┐  ┌──────────┐
                        │ PMCF Report │  │ CER Update? │
                        │(independent)│  └──────────┘
                        └──────────┘
```

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2. Data Model

2.1 Hierarchy

```
Project
├─ PMS Configuration
│  ├─ PMS Plan (document — strategy and methods)
│  ├─ PMCF Plan (document — clinical follow-up strategy)
│  ├─ Similar Device Registry (shared with SOA)
│  └─ Gap Registry (aggregated from SOA, Validation, CER)
│
└─ PMS Cycles (one per reporting period)
   ├─ Cycle Configuration (period, linked CER, linked SLS sessions)
   ├─ PMCF Activities
   │  ├─ Survey Results (imported)
   │  ├─ Literature Update (SLS pms_update session)
   │  ├─ Named Device Literature Update (SLS named_device session)
   │  └─ Vigilance Monitoring Results (competent authority searches)
   ├─ Complaints & Incidents Summary (manually entered from ZOHO Desk)
   ├─ Installed Base Data (manually entered)
   ├─ Maintenance Activity Summary (manually entered)
   ├─ Trend Analysis (computed from entered data)
   ├─ PMCF Report (generated document — independent)
   ├─ PSUR (generated document — comprehensive annual report)
   └─ CER Update Decision (documented conclusion)
```

2.2 PMS Configuration Record

Field	Description
id	UUID
project_id	Parent project
update_frequency	e.g., "1 year" (default for Class IIb under MDR)
first_psur_target_date	e.g., "March 2026"
cover_period_years	How many years of data each PSUR covers (e.g., 4 for initial, then rolling)
similar_devices	List of similar devices to monitor (shared with SOA similar device registry)
vigilance_databases	List of competent authority databases to search (configurable)
vigilance_keywords	Keywords per similar device for vigilance searches
survey_method	Description of survey approach
survey_distribution_method	e.g., "Sent to distributors based on installed base list"
survey_template_id	Link to the survey template
pms_plan_status	draft → approved → active
pmcf_plan_status	draft → approved → active

2.3 PMS Cycle Record

Field	Description
id	UUID
pms_config_id	Parent PMS configuration
cycle_number	Sequential (1, 2, 3...)
period_start	Start of reporting period
period_end	End of reporting period
linked_cer_id	Which CER version this cycle relates to
status	active → data_collection → analysis → report_draft → completed → locked
psur_version	PSUR document version

Field	Description
pmcf_report_version	PMCF Report document version
cer_update_decision	no_update_needed | update_triggered | pending
cer_update_justification	Text — why a CER update is or isn't needed
notes	Free-form

3. PMS Plan

3.1 Concept

The PMS Plan is a strategy document defining the post-market surveillance approach. It is generated once per project (with updates as needed) and follows the structure of the current PMS Plan template.

3.2 PMS Plan Section Template

```
1. INTRODUCTION
   ├── 1.1 Document Overview
   ├── 1.2 Abbreviations
   └── 1.3 Document References

2. SCOPE
   Source: project definition (device name, classification, market status,
   IMDRF SaMD category)
   Mode: auto-populated from project

3. RESPONSIBILITIES
   Source: manual / project team configuration
   Mode: editable table of responsibilities per PMS activity

4. COVER PERIOD
   Source: PMS configuration (cover_period_years)
   Mode: auto-populated

5. COLLECTED & ANALYZED DATA PERTAINING TO [DEVICE]
   │ This section defines each data source, responsibility,
   │ periodicity, processing method, and acceptance criteria.
   │
   ├── Maintenance Activities
   ├── Installed Base (Quantity of Licenses)
   ├── Complaints — Problems/Incidents
   ├── Complaints — Features/Questions
```

```
  ├── Software Anomalies (Non-conformities)
  ├── Serious Incidents (FSN/FSCA)
  ├── Surveys
  ├── Scientific Literature
  └── Monitoring of External Registries/Databases

  Each data source has a structured record:

  | Field | Description |
  |-------|-------------|
  | source_name | e.g., "Complaints — Problems/Incidents" |
  | responsibility | e.g., "Customer support agents" |
  | periodicity | e.g., "Monthly during CCC, yearly for PSUR" |
  | data_processing_method | e.g., "CRM Data (ZOHO Desk)" |
  | acceptance_criteria | e.g., "Trend analysis — significant increase defined as..." |
  | threshold_definition | e.g., ">10 points increase in incident ratio over one quarter" |

6. COLLECTED & ANALYZED DATA PERTAINING TO SIMILAR DEVICES
  ├── 6.1 Vigilance Data and Safety Measures
  │   Source: vigilance database list from PMS configuration
  │   Content: table of databases × keywords × periodicity
  │
  └── 6.2 Literature Review on Similar Devices
      Source: reference to SLS pms_update session
      Content: methodology description

7. POST-MARKET CLINICAL FOLLOW-UP ACTIVITIES
  Source: reference to PMCF Plan
  Mode: summary + link

8. EVALUATION OF POST-MARKET DATA AND UPDATE FREQUENCY
  Source: PMS configuration
  Content: PSUR schedule, what triggers a CER update
```

3.3 Generation

The PMS Plan is AI-drafted from the PMS configuration and project data. The user reviews and edits. Export as DOCX.

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4. PMCF Plan

4.1 Concept

The PMCF Plan is the clinical follow-up strategy — a companion to the PMS Plan focused specifically on clinical data collection. It follows MDCG 2020-7 template.

4.2 PMCF Plan Section Template

```
1. INTRODUCTION
  ├── 1.1 Document Overview
  └── 1.2 Abbreviations

2. MANUFACTURER CONTACT DETAILS
  Source: project definition
  Mode: auto-populated

3. MEDICAL DEVICE DESCRIPTION AND SPECIFICATION
  Source: project definition (same as CER §3.2)
  Mode: auto-populated table

4. ACTIVITIES RELATED TO PMCF
  │
  │ Master table summarizing all PMCF activities:
  │ | Activity | Implemented? | Methods | Aim | Timelines |
  │
  ├── 4.1 Surveys
  │   ├── 4.1.1 Risks — Safety (survey questions targeting
  │   │       risk monitoring)
  │   ├── 4.1.2 Performances (survey questions for Se/Sp
  │   │       estimation)
  │   └── 4.1.3 Misuse — Off-label Use
  │
  ├── 4.2 Systematic Literature Search
  │   Source: reference to SLS sessions (pms_update + named_device)
  │   Content: description of search methodology and databases
  │
  ├── 4.3 PMCF Studies and Clinical Investigations
  │   Content: justification for why no PMCF study is needed
  │   (or description of planned study if applicable)
  │
  ├── 4.4 External Registries
  │   Content: justification for why no registry exists for SaMD
  │
  └── 4.5 Manufacturer Registry
      Content: justification for why none implemented

5. REFERENCE TO RELEVANT PARTS OF THE TECHNICAL DOCUMENTATION
  ├── 5.1 Clinical Evaluation Report
  │   Source: linked CER → key findings to monitor
  │   Content: specific items from CER requiring PMCF monitoring
  │   (e.g., "vertebrae degenerative disease", "complex osteophytes")
  │
```

```
└── 5.2 Risk Management File
    Source: user-provided risk reference
    Content: specific risks requiring PMCF monitoring
    (e.g., "Risk #U4 — off-label use of viewer for
    primary interpretation")

6. EVALUATION OF CLINICAL DATA RELATED TO SIMILAR DEVICES
   Source: similar device registry
   Content: table with device name, intended purpose, users,
   population, condition, indication

7. APPLICABLE STANDARDS AND GUIDANCE
   Source: manual / standard library
   Content: list of applicable guidance documents
   (MEDDEV 2.12/2, MDCG 2020-6, 2020-7, 2020-1)

8. ESTIMATED DATE OF THE PMCF EVALUATION REPORT
   Source: PMS configuration
   Content: "PMCF report will be updated annually to feed
   the PSUR"
```

4.3 Gap Registry

The PMCF Plan is driven by identified gaps that need post-market monitoring. These gaps are aggregated from upstream modules:

Source	Gap Type	Example
SOA Clinical / Device	Evidence gaps	"Limited data on AI CSpine fracture detection in patients with ankylosing spondylitis"
SOA Device §5.2	Identified limitations	"No Canon scanner data in validation"
Validation — Fairness	Demographic disparities	"Performance disparity in age group ≥75 — PMCF action needed"
Validation — Labeling	Labeling gaps	"Not validated for pediatric use — monitor for off-label"
CER — Benefit-Risk	Residual risks	"Risk #U4 — off-label use requires monitoring"
CER — Discussion	Open questions	"Long-term performance stability to be confirmed"

Gap Record:

Field	Description
id	UUID
pms_config_id	Parent PMS configuration
source_module	soa_clinical | soa_device | validation | cer
source_reference	Specific section/item that identified this gap
gap_description	What is the gap
pmcf_action	What PMCF activity addresses this gap
pmcf_activity_type	survey | literature_update | pmcf_study | registry_monitoring | other
acceptance_criteria	When is this gap considered resolved
status	open | monitoring | resolved | accepted_limitation | carried_forward
resolution_summary	How the gap was resolved (if resolved)
cycle_id	Which PMS cycle addresses/resolved this gap

Gaps auto-feed into the PMCF Plan §4 (activities) and §5 (items to monitor). They also appear in the PSUR gap status summary. Gaps may persist across multiple PMS cycles — the carried_forward status tracks gaps that span multiple PSUR periods.

4.4 Survey Template Management

The PMCF Plan includes a survey template with structured questions. In CORTEX v1, the survey template is managed as a document (not an interactive form builder):

Field	Description
id	UUID
pms_config_id	Parent PMS configuration
name	e.g., "CINA-CSpine PMCF Survey v1"
sections	JSON — structured list of survey sections and questions
version	Template version
status	draft | approved

The survey template defines the questions but does not manage distribution or collection (v1). Results are imported separately (see §5.3).

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5. PMS Cycle — Data Collection

5.1 Concept

Each PMS cycle represents one reporting period (typically 1 year). During a cycle, various data sources are collected, analyzed, and assembled into the PMCF Report and PSUR.

5.2 Installed Base & Maintenance Summary

Manually entered structured data.

Installed Base Record:

Field	Description
cycle_id	Parent PMS cycle
total_licenses	Number of installed licenses
licenses_by_region	JSON: `{ "EU": N, "US": N, "Other": N }`
licenses_by_version	JSON: `{ "v1.0.0": N, "v1.0.1": N }`
estimated_usage_volume	If available — number of cases processed
data_source	e.g., "CRM Data (ZOHO)"
notes	e.g., "Due to B2B nature, exact usage frequency is estimated"

Maintenance Summary:

Field	Description
cycle_id	Parent PMS cycle
software_versions_on_market	List of versions currently deployed
patches_released	List of patches released during the period with descriptions
version_changes_summary	Text summary of what changed between versions

5.3 Survey Results (Imported)

In v1, survey results are imported as structured data (not collected through CORTEX).

Survey Import Record:

Field	Description
cycle_id	Parent PMS cycle
survey_template_id	Which survey template was used
import_date	When results were imported
total_surveys_sent	Number of surveys distributed
total_responses	Number of responses received
response_rate	Calculated percentage
results_data	JSON — structured responses per question
raw_file	Uploaded file (CSV/XLSX) with raw responses

Structured Results per Question Category:

For **Risk/Safety** questions:

Field	Description
question_key	e.g., "encountered_issue"
yes_count	Number of "Yes" responses
no_count	Number of "No" responses
details	Free text — aggregated "If yes, specify" responses

For **Performance** questions:

Field	Description
question_key	e.g., "false_positives_per_100"
responses	Array of numeric values
mean	Calculated mean
median	Calculated median
estimated_sensitivity	Computed from FN data
estimated_specificity	Computed from FP data
comparison_to_target	Comparison with Validation target performance

For **Misuse/Off-label** questions:

Field	Description
question_key	e.g., "off_label_use"
yes_count	Number of "Yes"
no_count	Number of "No"
details	Aggregated descriptions

5.4 Literature Updates

Two SLS sessions are executed (or re-executed) per PMS cycle:

1. **Similar Device Literature Update** — SLS type: `pms_update`
    - Re-executes the SOA device literature search with updated date range
    - Identifies new publications about the similar device
    - New articles are screened, included/excluded
    - Results feed into PSUR §8 and may trigger SOA update
2. **Named Device Literature Update** — SLS type: `named_device`
    - Re-executes the named device search with updated date range
    - Identifies any new publications about the device itself
    - Results feed into PSUR §8 and CER §5.3 update

Both use the full SLS workflow (query → execute → screen → PRISMA). The sessions are linked to the PMS cycle for traceability.

Literature Update Summary Record:

Field	Description
cycle_id	Parent PMS cycle
sls_session_id	Linked SLS session
search_type	similar_device | named_device
previous_sls_session_id	The SLS session from the previous cycle (for delta)
new_articles_found	Count of new articles identified
new_articles_included	Count of new articles included
new_articles_excluded	Count excluded (with reason summary)
key_findings_summary	Text — what the new literature says
impact_assessment	no_impact | minor_update | significant_update | cer_update_needed
impact_justification	Why the new literature does or doesn't change the conclusions

5.5 Vigilance Monitoring

Searches across competent authority databases for safety data about the named device and similar devices.

Vigilance Search Record:

Field	Description
cycle_id	Parent PMS cycle
database_name	e.g., "MAUDE (FDA — USA)"
country	e.g., "USA"
keywords_used	e.g., "AIDOC" AND "BriefCase"
search_date	When the search was performed
device_searched	named_device | similar_device
results_count	Number of hits
relevant_results	Number of relevant results after screening
findings_summary	Text — description of any findings
adverse_events_found	Boolean
fsn_fsca_found	Boolean
impact_assessment	no_impact | new_risk_identified | existing_risk_confirmed

Per-database list is pre-configured in PMS Configuration and matches the PMCF Plan §6.1 table. The user fills in results after each search.

5.6 Complaints & Incidents Summary

Manually entered structured summary of complaints data from ZOHO Desk CRM, classified using a dual-axis system aligned with Avicenna.AI's ticket management procedure (AV-I19-01) and IMDRF adverse event terminology (IMDRF/AE WG/N43FINAL:2023).

Axis 1 — Ticket Classification (from ZOHO Desk AV-I19-01 §4.7):

Classification	Definition	PMS Relevance
Cybersecurity Issue	Security issue, escalated to CISO per ISO 27001	Reported in PSUR if device-related
Duplicate	Same ticket opened twice	Excluded from PMS counts
Feature	Use outside intended use — request for new feature	Not a complaint. Tracked via AV-P03 Change Control
Spam	Commercial advertising	Excluded from PMS counts
Performance	False result during proper use, within acceptable range	Recorded as trend analysis. Core PMS data.
Question/Other	Cause other than device (price, info request)	Not device-related. Excluded from complaint counts
Problem/Incident	Confirmed complaint — failure, deficiency, misuse, or inadequacy	Core of PSUR §5 analysis
Serious Incident	Anomaly causing intolerable patient safety risk	Triggers AV-P17 Vigilance. PSUR §5.3

Axis 2 — Problem Category (IMDRF Code) (from AV-P17-APP3, based on IMDRF/AE WG/N43FINAL:2023):

Pre-loaded IMDRF codes relevant to SaMD, mapped to ZOHO Desk categories:

ZOHO Code	IMDRF Code	IMDRF Term
PERF - False Negative	A090803	False Negative Result
PERF - False Positive	A090804	False Positive Result
PERF - Incorrect Result	A0908	Incorrect, Inadequate or Imprecise Result
PERF - Delayed Execution	A110301	Delayed Program or Algorithm Execution
FUNC - Algorithm Failure	A110303	Program or Algorithm Execution Failure
FUNC - App Freezes	A110201	Application Program Freezes, Becomes Nonfunctional
FUNC - No Output	A0907	No Device Output
FUNC - Display Error	A090204	Image Display Error / Artifact
FUNC - Failure to Read Input	A13	Communication or Transmission Problem
SEC - Security Problem	A1105	Computer System Security Problem
LBL - Labelling/IFU	A2101	Device Markings / Labelling Problem
USB - Off-Label Use	A2304	Off-Label Use
USB - Improper Use	A2303	Improper or Incorrect Procedure or Method
USB - Output visual feedback	A0902	Display or visual feedback problem
TBD - To Be Defined	A27	Appropriate Term/Code Not Available

Priority field (High/Medium/Low): Not included in PMS data model. Priority is used exclusively for SLA management per customer contracts (AV-F08-04 Economic Operator's Communications) and customer satisfaction tracking. It is not relevant to PMS/PSUR analysis.

Complaints Summary Record:

Field	Description
cycle_id	Parent PMS cycle
period_covered	Date range
total_tickets	Total CRM tickets received
tickets_by_classification	JSON: breakdown by ZOHO Classification ({ "Problem/Incident": N, "Performance": N, "Serious Incident": N, "Feature": N, "Question/Other": N, "Cybersecurity Issue": N, "Duplicate": N, "Spam": N })
tickets_by_imdrf_code	JSON: breakdown by IMDRF Problem Category code ({ "A090803": N, "A090804": N, "A110303": N, ... })
complaints_count	Count of tickets classified as Problem/Incident + Serious Incident + Performance (= PMS-relevant complaints)
non_complaint_count	Count of Feature + Question/Other + Spam + Duplicate (= excluded from PMS analysis)
serious_incidents_count	Count of Serious Incident classification
serious_incidents_details	Structured list: date, description, IMDRF code, actions taken, notification status, risk matrix reference per AV-P07
incident_rate	Calculated: complaints_count / installed_base (see §5.7)
trend_tickets	Count of tickets with ZOHO Integration Status = "Trend analysis"
trend_details	Structured list of trend tickets with IMDRF codes, monitoring status, and CCC review outcome
nc_capa_summary	Non-conformities and CAPA summary for the period (ref: AV-F19-01 — List of nonconformities, List of CAPA)
notes	Free-form

Integration Status tracking: The ZOHO Desk Integration Status field tracks the lifecycle of complaint resolution:

Integration Status	Meaning
Trend analysis	Monitored — reviewed at each CCC and PSUR cycle
Patch TBP	Patch to be planned — issue requires software correction
Patch planned	Correction integrated in project team planning
Patch released	Application updated and deployed on market

During each PSUR cycle, all "Trend analysis" tickets are reviewed (per monthly CCC — Customer Complaint Committee process) to determine if escalation to "Patch TBP" is needed or if the issue presents a new risk or limitation.

5.7 Trend Analysis

The system computes trend indicators from entered data.

Incident Rate Formula:

> i = n / d (in %)
> where n = number of events in time interval t
> where d = used volume (installed base) in interval t

Significance Thresholds (configurable per project):

- High-volume products (>10,000 uses): rapid increase >10 points in one quarter
- Low-volume products (<10,000 uses): slow increase >10 points in one year

When a threshold is exceeded, the system:

1. Flags the cycle with a warning
2. Requires documentation of the investigation
3. Logs the event for PSUR inclusion
4. Documents whether EUDAMED notification may be required (per applicable QMS procedure)

Trend Visualization:

The system displays:

- Monthly incident counts over time (bar chart)
- Incident rate trend line
- Threshold line overlay
- Breakdown by IMDRF code category (stacked)
- Breakdown by Classification (Problem/Incident vs. Performance vs. Serious Incident)

6. PMCF Report

6.1 Concept

The PMCF Report aggregates all clinical follow-up activities from a PMS cycle and presents the findings. It is an **independent document** — produced before each PSUR and referenced by (or included in) the PSUR as input. It is updated for each PSUR cycle.

6.2 PMCF Report Section Template

```
1. INTRODUCTION
   ├── 1.1 Document Overview
   ├── 1.2 Scope and Period Covered
   └── 1.3 Summary of PMCF Activities Performed

2. SURVEY RESULTS
   ├── 2.1 Survey Distribution and Response Rate
   ├── 2.2 Safety/Risk Findings
   ├── 2.3 Performance Assessment
   │   └── Comparison to CER target performance (Se, Sp)
   ├── 2.4 Misuse / Off-Label Use Assessment
   └── 2.5 Open Feedback Analysis

3. LITERATURE UPDATE — SIMILAR DEVICES
   ├── 3.1 Search Strategy (reference to SLS session)
   ├── 3.2 New Articles Found
   ├── 3.3 Key Findings
   └── 3.4 Impact Assessment on Benefit-Risk

4. LITERATURE UPDATE — NAMED DEVICE
   ├── 4.1 Search Strategy
   ├── 4.2 Results
   └── 4.3 Impact Assessment

5. VIGILANCE MONITORING
   ├── 5.1 Search Results per Database
   ├── 5.2 Findings Summary
   └── 5.3 Impact Assessment

6. GAP REGISTRY STATUS UPDATE
   ├── 6.1 Gaps Resolved
   ├── 6.2 Gaps Still Open
   ├── 6.3 Gaps Carried Forward from Previous Cycles
   └── 6.4 New Gaps Identified
```

```
7. CONCLUSIONS
  ├── 7.1 Safety Confirmed / Concerns Identified
  ├── 7.2 Performance Confirmed / Concerns Identified
  ├── 7.3 Benefit-Risk Ratio Assessment
  └── 7.4 Recommendations (PMCF Plan update needed? New actions?)

8. REFERENCES
```

6.3 Generation

The PMCF Report is AI-drafted from the cycle's collected data:

- Survey results → §2
- SLS pms_update session → §3
- SLS named_device session → §4
- Vigilance search records → §5
- Gap registry → §6
- Conclusions synthesized from all above

Each section is reviewable and editable. Export as DOCX.

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7. PSUR (Periodic Safety Update Report)

7.1 Concept

The PSUR is the comprehensive annual report required by MDR Article 86. For Class IIa/IIb/III devices, it aggregates ALL post-market data and concludes with a benefit-risk re-assessment.

The PSUR is the **sole comprehensive post-market report** under MDR. It includes both non-clinical data (complaints, incidents, maintenance, installed base) and clinical data (from the PMCF Report). There is no separate "PMS Report."

7.2 PSUR Section Template

```
1. INTRODUCTION
  ├── 1.1 Document Overview
  ├── 1.2 Abbreviations
  ├── 1.3 Document References
  └── 1.4 Device Description (auto-populated from project)

2. SCOPE AND PERIOD
  ├── 2.1 Reporting Period
  ├── 2.2 Previous PSUR Reference (if applicable)
```

└── 2.3 Device Versions Covered

3. INSTALLED BASE AND USAGE DATA
Source: installed base record
Content: licenses by region, version, estimated usage

4. MAINTENANCE ACTIVITIES
Source: maintenance summary record
Content: versions on market, patches released, changes

5. COMPLAINTS AND INCIDENTS — NAMED DEVICE
├── 5.1 Complaint Summary
│ Source: complaints summary record
│ Content: total tickets, breakdown by Classification
│ (Problem/Incident, Performance, Serious Incident),
│ breakdown by IMDRF Problem Category code,
│ complaint vs. non-complaint separation, incident rate
│
├── 5.2 Trend Analysis
│ Source: trend analysis data
│ Content: graphical trend representation (incident rate
│ over time, by IMDRF code, by Classification),
│ threshold assessment, CCC review outcomes
│
├── 5.3 Serious Incidents
│ Source: complaints summary → serious incidents details
│ Content: each incident with description, IMDRF code,
│ actions taken, notification status,
│ risk matrix reference per AV-P07
│
└── 5.4 Non-Conformities and CAPA
Source: complaints summary → nc_capa_summary
Content: summary of non-conformities and corrective
actions for the period
(ref: AV-F19-01 — List of nonconformities, List of CAPA)

6. VIGILANCE DATA — SIMILAR DEVICES
├── 6.1 Named Device — Competent Authority Search Results
│ Source: vigilance search records (named device)
│ Content: per-database results table
│
└── 6.2 Similar Device — Competent Authority Search Results
Source: vigilance search records (similar device)
Content: per-database results table + findings analysis

7. POST-MARKET CLINICAL FOLLOW-UP (PMCF)
Source: PMCF Report (summary or inline reference)

```
  Content: survey results, literature updates, clinical findings
  Note: references the full PMCF Report as an appendix
  or includes key sections inline

8. SCIENTIFIC LITERATURE UPDATE
  ├── 8.1 Similar Device Literature
  │   Source: SLS pms_update session → summary
  │   Content: new articles, key findings, PRISMA delta
  │
  └── 8.2 Named Device Literature
      Source: SLS named_device session → summary
      Content: new articles found (or "none")

9. REGULATORY AND NORMATIVE DATA
  Content: applicable standards, guidance documents,
  regulatory changes during the period
  Mode: manual / standard library

10. BENEFIT-RISK RE-ASSESSMENT
  ├── 10.1 Current Benefit Assessment
  │   Source: CER benefit registry (current) + any new
  │   evidence from PMCF
  │   Content: are benefits confirmed by post-market data?
  │
  ├── 10.2 Current Risk Assessment
  │   Source: complaints/incidents analysis + vigilance
  │   + literature
  │   Content: are risks unchanged? new risks identified?
  │
  └── 10.3 Benefit-Risk Conclusion
      Content: "The benefit-risk ratio remains acceptable"
      or "Concerns identified — CER update needed"

11. CONCLUSIONS AND ACTIONS
  ├── 11.1 Overall Conclusion
  ├── 11.2 Need for Corrective Actions / FSCA
  ├── 11.3 Need for Benefit-Risk / Risk Assessment Update
  ├── 11.4 Need for Clinical Evaluation Update (CER)
  ├── 11.5 Need for PMS Plan / PMCF Plan Update
  ├── 11.6 Need for Design / IFU / Labeling Update
  └── 11.7 Next PSUR Date

12. APPENDICES
  ├── PMCF Report (full — independent document)
  ├── Literature Search Reports (SLS exports)
  └── Vigilance Search Details
```

7.3 CER Update Decision

Section §11.4 of the PSUR formally documents the decision on whether a CER update is needed. The decision is stored in the PMS cycle record (`cer_update_decision`) with justification.

Decision criteria (auto-suggested by the system):

Trigger	Suggested Decision
New significant literature changes SOA conclusions	`update_triggered`
New adverse events or safety signals identified	`update_triggered`
Incident rate exceeds threshold	`update_triggered`
Performance data from surveys differs significantly from CER claims	`update_triggered`
New similar device identified	`update_triggered`
No new data, all indicators stable	`no_update_needed`
New patch released with re-validation	`update_triggered` (patch CER update)

The user confirms or overrides the suggestion. If `update_triggered`, the system offers to create a new CER version linked to this PMS cycle.

7.4 Generation

The PSUR is AI-drafted from all cycle data:

- Installed base + maintenance → §3-4
- Complaints/incidents (by Classification and IMDRF code) → §5
- Vigilance searches → §6
- PMCF Report data → §7
- Literature update SLS sessions → §8
- Benefit-risk re-assessment → §10

Each section is reviewable and editable. Export as DOCX.

---

8. FDA Post-Market Requirements

8.1 Scope

The PMS module is primarily designed for MDR Article 83-92 compliance. FDA post-market requirements

differ in several areas and are documented here for completeness and future implementation.

8.2 Known Gaps

Area	MDR Requirement	FDA Equivalent	Status in CORTEX
Annual reporting	PSUR (Article 86)	Annual Report (21 CFR §814.84, PMA only)	PSUR template covers MDR. FDA Annual Report template deferred.
Postmarket surveillance studies	PMCF (MDCG 2020-7)	522 Postmarket Surveillance Studies (21 USC §360l)	Not covered in v1. FDA may require specific studies as condition of clearance.
Adverse event reporting	Vigilance (Article 87-92)	MDR — Medical Device Reporting (21 CFR Part 803)	Complaint classification (IMDRF codes) shared. Reporting forms (MedWatch 3500A) and timelines differ — managed via QMS.
Trend reporting	Trend analysis in PSUR	21 CFR §803.53 (MDR trend analysis)	Trend analysis methodology shared. FDA-specific thresholds may differ.
Periodic reporting	PSUR annually	FDA: varies by submission type	Not applicable for 510(k). PMA requires annual report.

Part 522 reference: FDA 522 Postmarket Surveillance Studies have their own protocol/reporting structure distinct from PMCF. If required as a condition of clearance, CORTEX should support a 522 Study tracking mode in a future version.

These gaps are documented and accepted. FDA-specific post-market deliverables will be added as additional export templates when needed. All timings and reporting obligations are managed through Avicenna.AI's QMS procedures, not enforced by CORTEX.

---

9. Workflow & Statuses

9.1 PMS Plan Lifecycle

The PMS Plan is approved once and updated only when the strategy changes. Updates create a new version.

```
draft → approved → active
```

9.2 PMCF Plan Lifecycle

Same as PMS Plan. Updated when new PMCF activities are added or methodology changes.

draft → approved → active

9.3 PMS Cycle Lifecycle

active → data_collection → analysis → report_draft → completed → locked

Status	What Happens	What's Editable
active	Cycle open, data collection ongoing	Everything
data_collection	Actively collecting: surveys sent, literature searches running, vigilance searches performed	Data entry, SLS sessions
analysis	All data collected, analyzing trends and writing summaries	Analysis fields, summaries
report_draft	PMCF Report and PSUR being drafted	Report content
completed	Reports finalized	Minor edits (logged)
locked	Cycle frozen. Read-only.	Nothing

10. Exports

Export	Format	Content
PMS Plan	DOCX	Full PMS strategy document
PMCF Plan	DOCX	Full PMCF strategy document including survey template
PMCF Report	DOCX	Clinical follow-up results for the cycle (independent document)
PSUR	DOCX	Full periodic safety update report (comprehensive annual report)
Gap Registry	XLSX	All gaps with status, source, action, resolution, carried-forward tracking
Survey Results Summary	XLSX	Structured survey results with analysis
Vigilance Search Summary	XLSX	Per-database vigilance search results
Trend Analysis Report	XLSX / PDF	Incident rates over time with graphical trends, by IMDRF code and Classification

Export	Format	Content
Literature Update Summary	XLSX	New articles found per SLS session with impact assessment
Complaints Summary	XLSX	Full breakdown by Classification and IMDRF code, trend tickets, NC/CAPA

11. Async Operations

Operation	Estimated Duration	Async?
PMCF Report generation (AI draft)	30-60s	Yes
PSUR generation (AI draft)	60-120s	Yes
PMS Plan generation (AI draft)	15-30s	No (inline)
PMCF Plan generation (AI draft)	15-30s	No (inline)
Trend analysis computation	< 2s	No (inline)
Survey import & analysis	5-15s	No (inline)
Report export (DOCX)	10-30s	No (inline with spinner)

12. Permissions

Role	Can Do
clinical / regulatory	Full access: manage PMS config, collect data, draft reports, analyze
manager	Full access + approve PMS/PMCF Plans, lock cycles, make CER update decisions
collaborateur	Read-only access to completed/locked cycles and exports
admin	Everything + manage survey templates, vigilance database list at system level

13. Open Items

1. **Survey builder (v2):** Currently surveys are managed as templates and results are imported. A future

version could include a survey builder with distribution via email/link, response collection, and automatic analysis — reducing the dependency on external tools.

2. **ZOHO CRM integration (v2):** Complaints and incidents data is manually entered in v1. API integration with ZOHO Desk and ZOHO CRM would automate import of ticket counts, categories (both Classification and IMDRF code), Integration Status, and trend data. This would eliminate manual data entry for §5.6 and improve data accuracy.
3. **EUDAMED integration:** When incident thresholds are exceeded, EUDAMED notification may be required. The system flags this but does not automate the EUDAMED submission. The notification decision and timing are managed per QMS procedures. Future integration with EUDAMED APIs could streamline this.
4. **Real-world performance monitoring:** The survey-based approach provides subjective performance estimates. A future enhancement could integrate anonymized processing logs from deployed instances (with consent) to compute objective sensitivity/specificity from real-world data.
5. **Multi-device PSUR:** A manufacturer with multiple similar devices (CINA-CSpine, CINA-ASPECTS, CINA-ICH) may want a consolidated PSUR or at least shared PMS infrastructure (same vigilance database list, shared similar device registry). The system should support shared PMS configurations with per-device overrides.
6. **Gap lifecycle across cycles:** Gaps may persist across multiple PMS cycles. The `carried_forward` status tracks this. A visual timeline of gap status across cycles would improve traceability for Notified Body reviews.
7. **PMCF Study management:** The spec currently supports "no PMCF study" (which is appropriate for most SaMD). If a PMCF study (observational) is needed in the future, the system would need a lightweight clinical study management module. This is deferred to v2.
8. **PSUR template variants — FDA:** The PSUR template follows MDR Article 86 structure. For FDA Annual Reports (21 CFR §814.84 for PMA), a different template is needed. The system should support regulatory-context-specific post-market report templates.
9. **Automatic benefit-risk re-assessment:** The PSUR §10 currently relies on the user to re-assess benefit-risk. A future enhancement could auto-compare: survey performance estimates vs. CER target performance, incident rates vs. baseline, new literature findings vs. current SOA conclusions — and flag discrepancies automatically.
10. **PMS Plan / PMCF Plan version control:** Currently these are single documents with manual versioning. The system should track version history with diff capability, similar to how CER versions are managed.
11. **IMDRF code list maintenance:** The IMDRF AE WG/N43 code list is periodically updated. CORTEX should support versioned code lists with backward compatibility — existing complaints retain their original code, new complaints use the updated list. A mapping table between code versions may be needed.
12. **CCC integration:** The monthly Customer Complaint Committee (CCC) process reviews trend analysis tickets. CORTEX could provide a dedicated CCC review view showing all "Trend analysis" tickets with

their IMDRF codes, recommended actions, and a structured decision record (escalate to Patch TBP / maintain as trend / close). CCC meeting reports (per AV-F19-01) could be linked to the PMS cycle.

13. **FDA Post-Market Requirements divergence (§8):** 522 Postmarket Surveillance Studies, MedWatch 3500A reporting, and FDA Annual Reports are documented gaps. FDA-specific templates and tracking modes will be added when needed.

Version History

Version	Date	Changes
v1.0	January 2026	Initial specification
v1.1	February 2026	Regulatory gap analysis, vigilance workflows, CUSUM trend detection, dual EU/US support framework
v1.2	February 2026	Internal review integration: SSCP removal, PMS Report removal (PSUR-only under MDR), dual-axis complaint classification (ZOHO Classification + IMDRF codes), Priority exclusion from PMS, Integration Status tracking, CAPA in PSUR confirmed, QMS timing delegation, FDA gaps documented with Part 522 reference, gap carried_forward status, RWE scope clarification, PMCF Report independence confirmed